

Ralph Jayrell E. Gacusan

Quezon City, Philippines • ralphjayrell.e.gacusan@gmail.com • 09333804710 • github.com/ralphgacusan • <https://ralphgacusan.vercel.app>

Education

Technological Institute of the Philippines
Bachelor of Science in Information Technology

Quezon City, Philippines
August 2023 – June 2027

Projects

Nurtura – Caregiving Coordination Backend

Python, FastAPI, PostgreSQL

- Engineered a scalable backend using FastAPI, PostgreSQL, SQLAlchemy, Alembic, JWT, and bcrypt, designing a modular architecture across 6 core modules and delivering secure RESTful APIs for collaborative care coordination.
- Implemented WebSockets, APScheduler, and the Gemini AI API, then containerized with Docker and deployed on Render, enabling real-time communication and seamless integration with a React Native mobile client.

Timplato – Kitchenware E-Commerce Website

PHP, Laravel, MySQL, JavaScript

- Developed a full-stack Laravel e-commerce platform with RBAC, secure authentication, and full CRUD functionalities for products, orders, users, inventory, and content, powering 6 core shopping features.
- Integrated Google OAuth, PayMongo, and Tawk.to APIs, then deployed on Hostinger, enabling secure authentication, online payments, and AI-assisted customer support.

Lecture-to-Document Web Application

Python, FastAPI, Google Vision API, Firebase

- Engineered an edge-to-cloud lecture digitization system where a Raspberry Pi device captured and preprocessed lecture images before transmitting them to a FastAPI backend that used the Google Vision API for OCR, enabling export to editable DOCX and PDF documents.
- Developed RESTful APIs supporting batch image uploads, Google Vision API OCR with automatic PyTesseract fallback, Firebase-based real-time device monitoring and control, and document generation to improve scalability, reliability, and fault tolerance.

Household Segmentation & KADIWA Eligibility Prediction

Python, Machine Learning

- Engineered an end-to-end machine learning pipeline in Python (Pandas, Scikit-learn, NumPy), preprocessing 41,554 household records, engineering 30+ features, and integrating PSA agricultural datasets to improve data quality and model readiness.
- Built and evaluated Random Forest and K-Means models, achieving 91.72% accuracy and 0.972 ROC-AUC, identifying 4 behavioral clusters using PCA, silhouette, and inertia evaluation to deliver data-driven recommendations for KADIWA targeting.

Certifications

CompTIA IT Fundamentals (ITF+) Certification

January 24, 2024

Test of Practical Competency in IT (Competent Level)

December 10, 2025

Cognitive Class – Data Analysis with Python

March 9, 2026

Achievements / Honors

Consistent Lister (1st Year, 1st Semester – 3rd Year, 2nd Semester)

Finalist - UP SoCompSci Tech Week 2026: Hack-it-UP Hackathon

May 16, 2026

Top 30 - eGovPH Hackathon

July 21 - 22, 2026

Technical Skills

Programming Languages: Python, Java, C#, PHP, JavaScript, R, SQL, HTML, CSS

Frameworks & Libraries: FastAPI, Laravel, Bootstrap, Tailwind CSS, daisyUI, jQuery

Databases: PostgreSQL, MySQL, Firebase

Development Tools & Environments: VS Code, Eclipse, Git, Jupyter Notebook, Docker, Postman, Render, RStudio

Data Analysis Tools: Excel, WEKA, RapidMiner

APIs & Integrations: RESTful APIs, Google OAuth 2.0, PayMongo API, Gemini AI API, Google Vision API